EX NO. : 9	INTRUSION DETECTION SYSTEM (IDS)
DATE :	

AIM:

To install and configure Snort on Ubuntu and demonstrate intrusion detection by identifying ICMP (ping) traffic.

OBJECTIVE:

- Understand signature-based Intrusion Detection Systems (IDS)
- Create and apply a custom Snort rule
- Generate real-time alerts for network traffic

REQUIREMENTS:

- Ubuntu Operating System
- Snort IDS
- Internet connection (only for installation)

PROCEDURE:

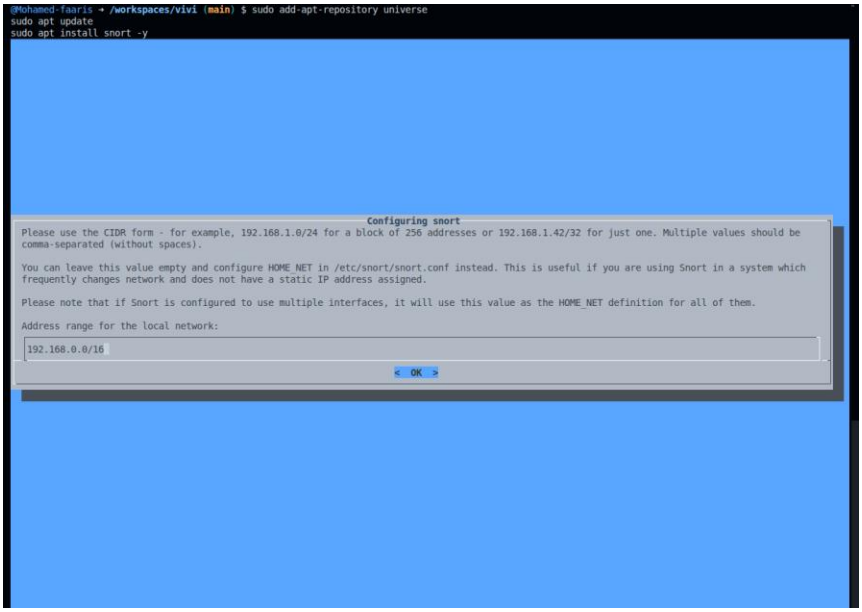
1. Install Snort

```
sudo add-apt-repository universe

sudo apt update

sudo apt install snort -y

sudo apt install iputils-ping -y
```



Verify installation:

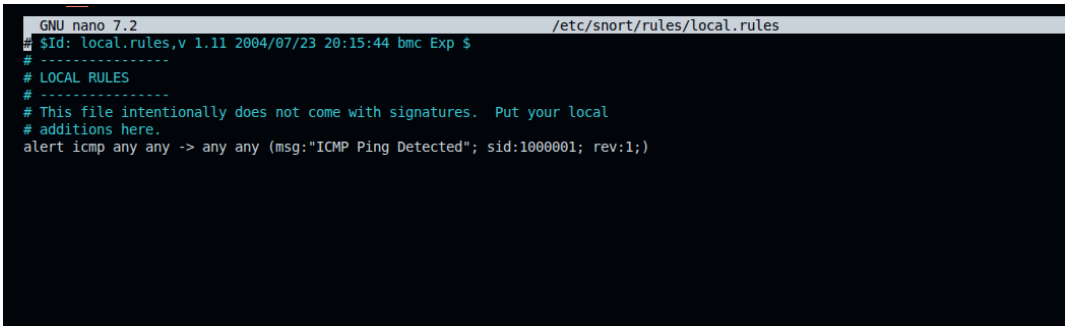
```
snort -V
```

3. Create Custom Rule

```
sudo nano /etc/snort/rules/local.rules
```

Add:

```
alert icmp any any -> any any (msg:"ICMP Ping
Detected"; sid:1000001; rev:1;)
```



4. Enable Rule in Configuration

```
sudo nano /etc/snort/snort.conf
```

Uncomment (probably already done):

```
include $RULE_PATH/local.rules
```

```
# If you install the official VRT Sourcefire rules please review this
# configuration file and re-enable (remove the comment in the first line) those
# rules files that are available in your system (in the /etc/snort/rules
# directory)

# site specific rules
include $RULE_PATH/local.rules

# The include files commented below have been disabled
# because they are not available in the stock Debian
# rules. If you install the Sourcefire VRT please make
# sure you re-enable them again:

#include $RULE_PATH/app-detect.rules
include $RULE_PATH/attack-responses.rules
include $RULE_PATH/backdoor.rules
```

5. Run Snort in Detection Mode

a. Identify Network Interface

```
ip a
```

```
● @Mohamed-faaris → /workspaces/vivi (main) $ ip a
 1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
     link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
     inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
     inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
 2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
     link/ether 7c:ed:8d:ba:57:15 brd ff:ff:ff:ff:ff:ff
     inet 10.0.1.213/16 metric 100 brd 10.0.255.255 scope global eth0
        valid_lft forever preferred_lft forever
     inet6 fe80::7eed:8dff:feba:5715/64 scope link
        valid_lft forever preferred_lft forever
 3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
     link/ether 02:42:6b:04:21:3d brd ff:ff:ff:ff:ff:ff
     inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever
○ @Mohamed-faaris → /workspaces/vivi (main) $
```

Note the active interface (e.g., `wlan0` or `eth0`).

b. Run snort

Use the identified interface to start Snort in detection mode:

```
sudo snort-i <interface_name>-A console-c
/etc/snort/snort.conf
```

Example:

```
sudo snort -i eth0 -A console -c /etc/snort/snort.conf
```

```
        --== Initialization Complete ==--

   ,,_     -*> Snort! <*-
  o"  )~   Version 2.9.20 GRE (Build 82)
   ''''    By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
           Copyright (C) 2014-2022 Cisco and/or its affiliates. All rights reserved.
           Copyright (C) 1998-2013 Sourcefire, Inc., et al.
           Using libpcap version 1.10.4 (with TPACKET_V3)
           Using PCRE version: 8.39 2016-06-14
           Using ZLIB version: 1.3

           Rules Engine: SF_SNORT_DETECTION_ENGINE  Version 3.2  <Build 1>
           Preprocessor Object: SF_DNP3  Version 1.1  <Build 1>
           Preprocessor Object: SF_SSLPP  Version 1.1  <Build 4>
           Preprocessor Object: SF_S7COMMPLUS  Version 1.0  <Build 1>
           Preprocessor Object: SF_MODBUS  Version 1.1  <Build 1>
           Preprocessor Object: SF_SSH  Version 1.1  <Build 3>
           Preprocessor Object: SF_IMAP  Version 1.0  <Build 1>
           Preprocessor Object: SF_SIP  Version 1.1  <Build 1>
           Preprocessor Object: SF_DNS  Version 1.1  <Build 4>
           Preprocessor Object: SF_REPUTATION  Version 1.1  <Build 1>
           Preprocessor Object: SF_GTP  Version 1.1  <Build 1>
           Preprocessor Object: SF_FTPTELNET  Version 1.2  <Build 13>
           Preprocessor Object: SF_DCERPC2  Version 1.0  <Build 3>
           Preprocessor Object: SF_SDF  Version 1.1  <Build 1>
           Preprocessor Object: appid  Version 1.1  <Build 5>
           Preprocessor Object: SF_POP  Version 1.0  <Build 1>
           Preprocessor Object: SF_SMTP  Version 1.1  <Build 9>
Commencing packet processing (pid=23810)
```

6. Generate Network Traffic

```
Use different terminal

ping 8.8.8.8
```

```
@Mohamed-faaris → /workspaces/vivi (main) $ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
```

OUTPUT

```
Commencing packet processing (pid=24970)
04/18-13:59:20.661111  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.0.1.213 -> 8.8.8.8
04/18-13:59:21.685080  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.0.1.213 -> 8.8.8.8
04/18-13:59:22.709197  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.0.1.213 -> 8.8.8.8
04/18-13:59:23.733112  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.0.1.213 -> 8.8.8.8
04/18-13:59:24.757120  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.0.1.213 -> 8.8.8.8
04/18-13:59:25.781119  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.0.1.213 -> 8.8.8.8
04/18-13:59:26.805095  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.0.1.213 -> 8.8.8.8
04/18-13:59:27.829098  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.0.1.213 -> 8.8.8.8
04/18-13:59:28.853124  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.0.1.213 -> 8.8.8.8
04/18-13:59:29.877119  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.0.1.213 -> 8.8.8.8
04/18-13:59:30.901062  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.0.1.213 -> 8.8.8.8
04/18-13:59:31.925085  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.0.1.213 -> 8.8.8.8
04/18-13:59:32.949098  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.0.1.213 -> 8.8.8.8
04/18-13:59:33.973116  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.0.1.213 -> 8.8.8.8
        04/18-13:59:34.997130  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.0.1.213 -> 8.8.8.8
04/18-13:59:36.021076  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.0.1.213 -> 8.8.8.8
04/18-13:59:37.045057  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.0.1.213 -> 8.8.8.8
04/18-13:59:38.069083  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.0.1.213 -> 8.8.8.8
04/18-13:59:39.093115  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.0.1.213 -> 8.8.8.8
04/18-13:59:40.117045  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.0.1.213 -> 8.8.8.8
04/18-13:59:41.141066  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.0.1.213 -> 8.8.8.8
04/18-13:59:42.165116  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.0.1.213 -> 8.8.8.8
04/18-13:59:43.189090  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.0.1.213 -> 8.8.8.8
04/18-13:59:44.213089  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.0.1.213 -> 8.8.8.8
04/18-13:59:45.237067  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.0.1.213 -> 8.8.8.8
04/18-13:59:46.261069  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.0.1.213 -> 8.8.8.8
04/18-13:59:47.285065  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.0.1.213 -> 8.8.8.8
04/18-13:59:48.309084  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.0.1.213 -> 8.8.8.8
04/18-13:59:49.333109  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.0.1.213 -> 8.8.8.8
04/18-13:59:50.357101  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.0.1.213 -> 8.8.8.8
04/18-13:59:51.381098  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.0.1.213 -> 8.8.8.8
04/18-13:59:52.405149  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.0.1.213 -> 8.8.8.8
04/18-13:59:53.429088  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.0.1.213 -> 8.8.8.8
04/18-13:59:54.453081  [**] [1:1000001:1] ICMP Ping Detected [**] [Priority: 0] {ICMP} 10.0.1.213 -> 8.8.8.8
```

[**] [1:1000001:1] ICMP Ping Detected [**]

RESULT:

Snort successfully detected ICMP traffic and generated real-time alerts. Thus, the experiment proves that Snort functions as a signature-based Intrusion Detection System (IDS) by comparing network traffic with predefined rules and successfully identifying ICMP packets in real time.

EX NO. : 10	**EXPLORING A VULNERABILITY ASSESSMENT TOOL**
DATE :	

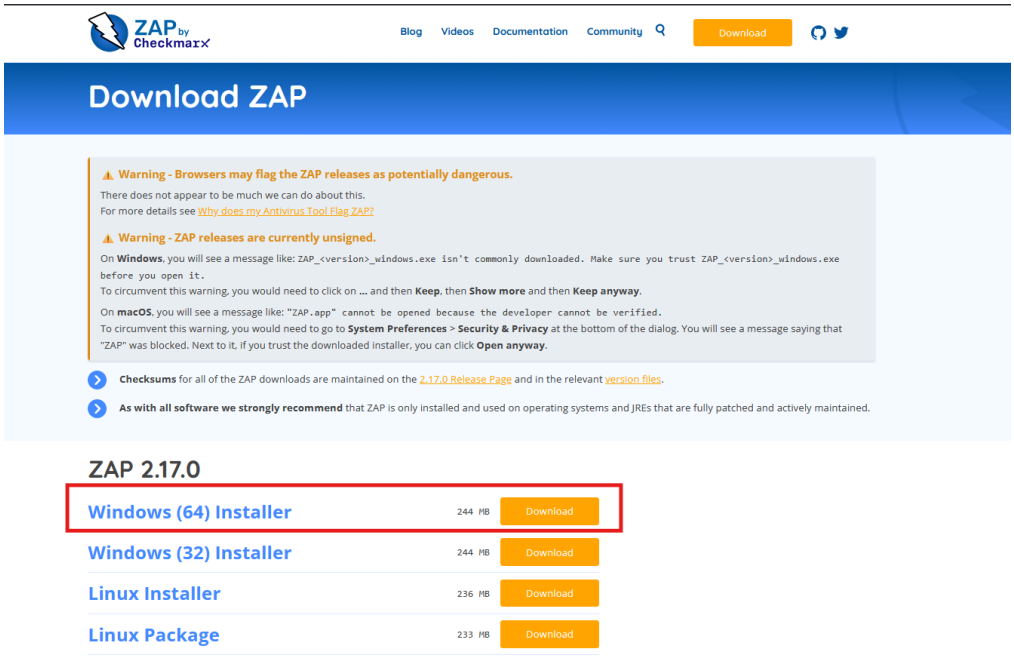
AIM:

To understand the functional requirements of a Vulnerability Assessment (VA) tool and to explore its features by scanning a target web application for security weaknesses.

PROCEDURE:

Step 1: Install ZAP from web browser.

URL: https://www.zaproxy.org/download/

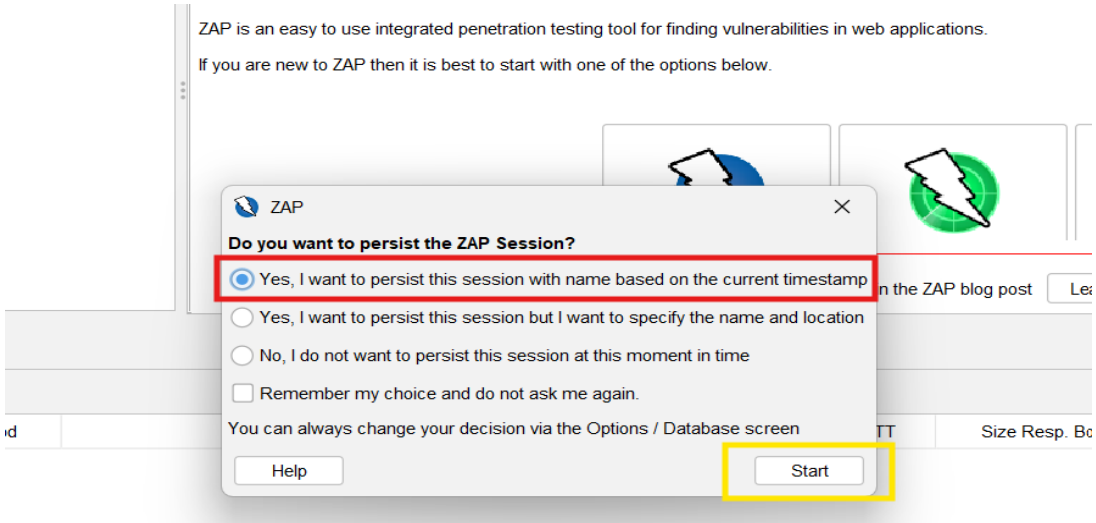


Step 2: Disable your antivirus if it's interrupting your installation of ZAP Scanner.

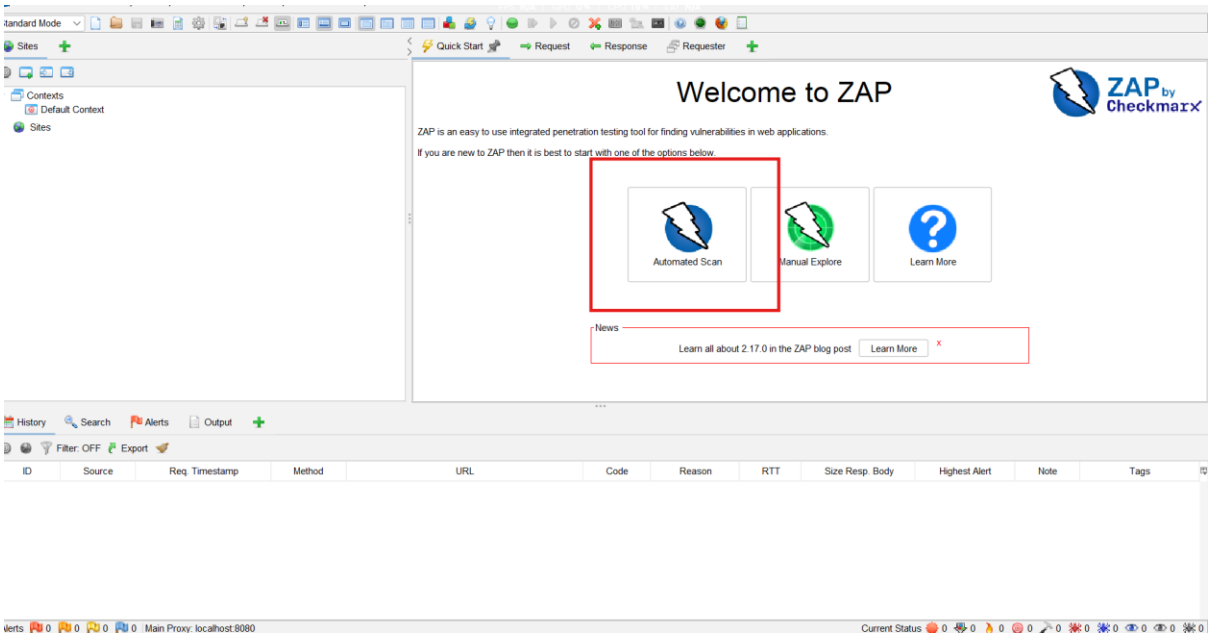
Step 3: Run the executable. A popup will appear, click "Yes". Click next for every step and click install.

Step 4: After installing the software, a reboot is required to efficiently run the ZAP Scanner.

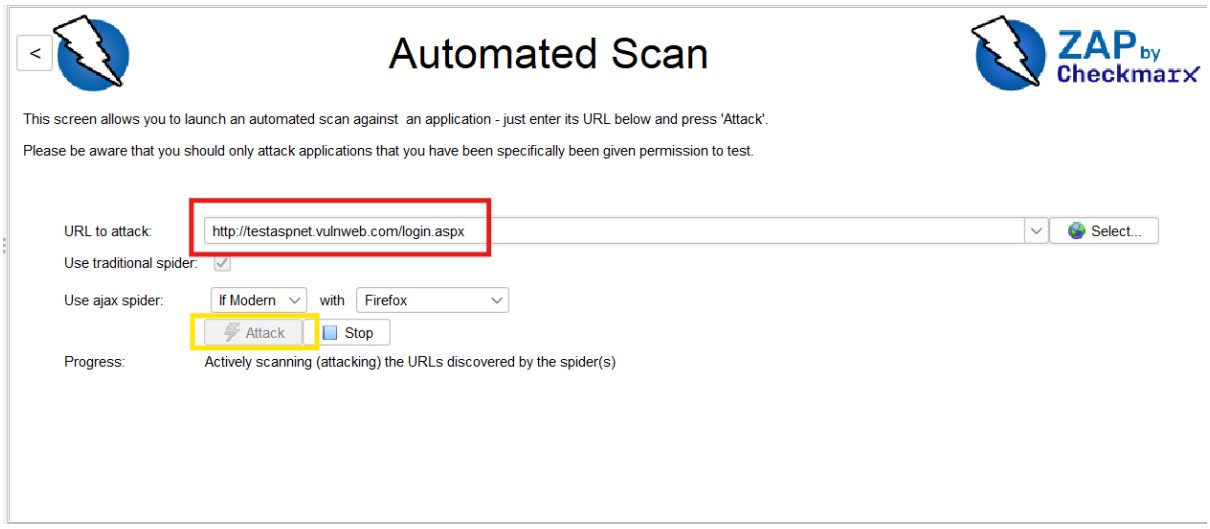
Step 5: Open the ZAP Scanner.



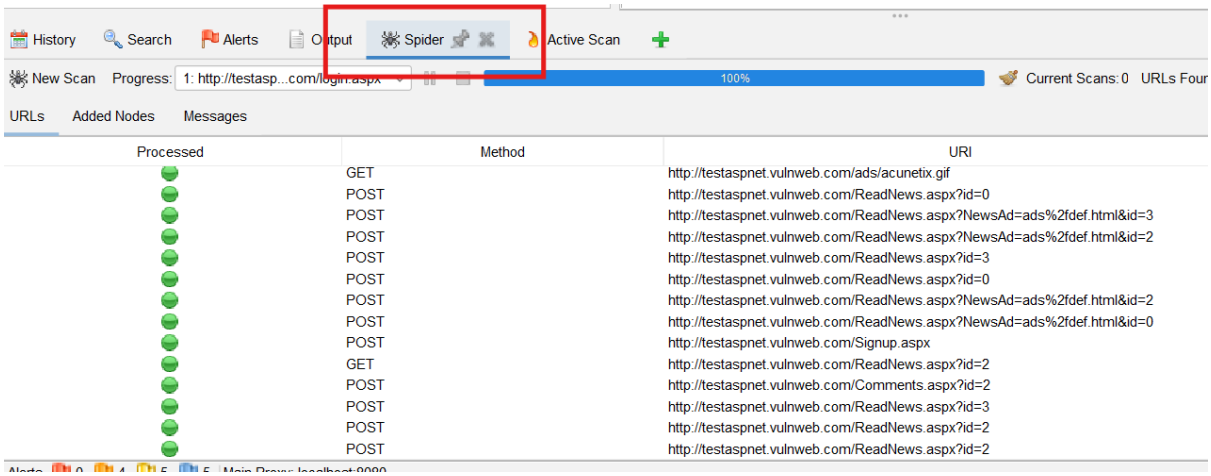
Step 6: Click the automated scan for scanning all type of vulnerabilities from the database.



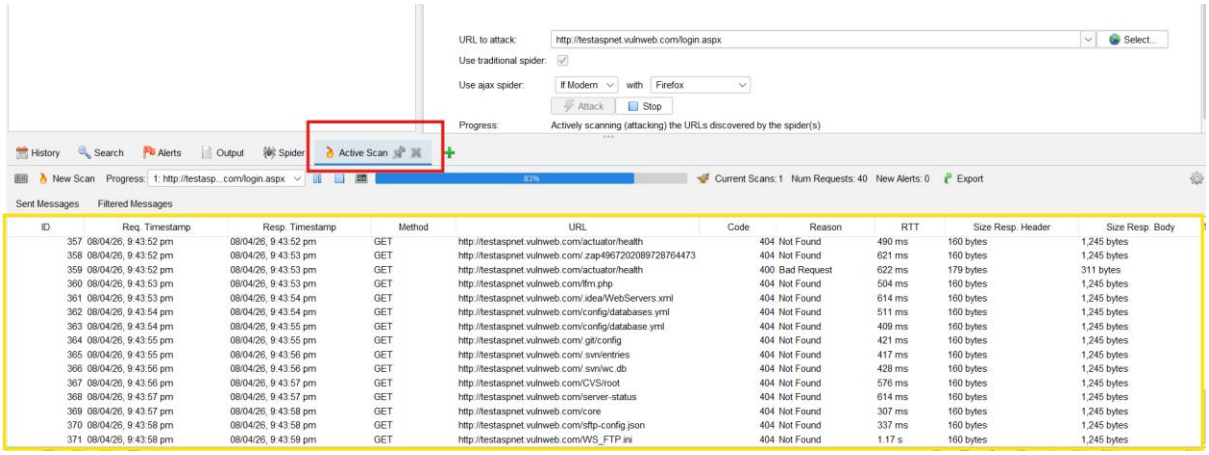
Step 7: Enter the URL in which you have to do the penetration testing (vulnerability testing) and enable "Use traditional spider" and click the **"Attack"** button.



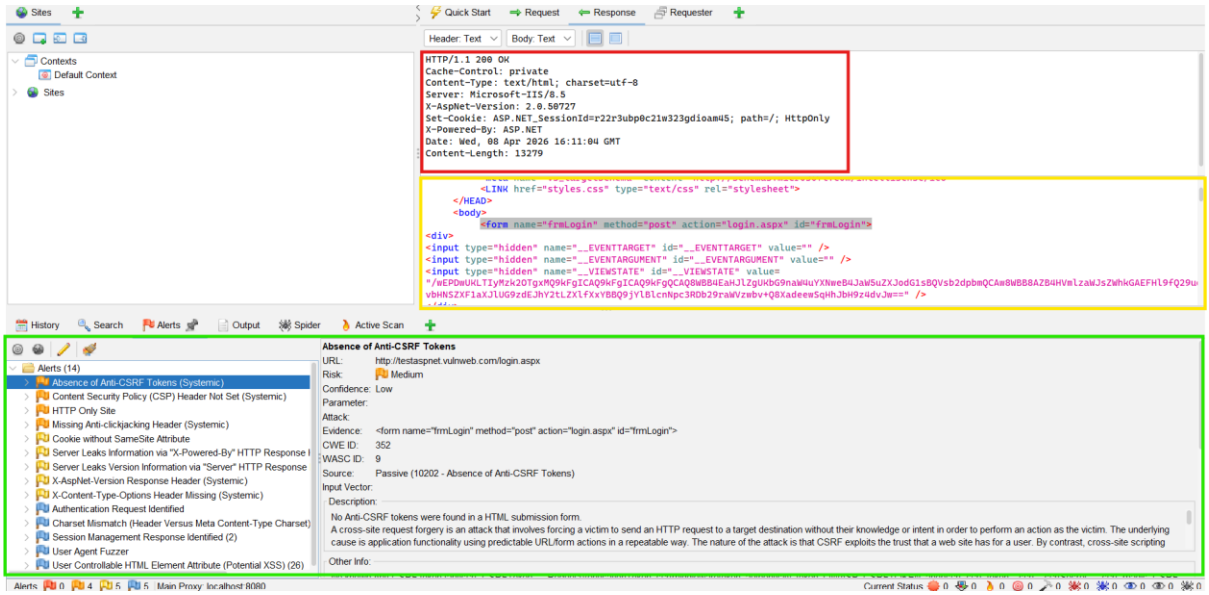
Step 8: The payloads will be tested one by one where we can notice immediately in the **"Spider Tab"**



Step 9: The Active scan will be started once the all available endpoints are scanned.



Step 10: Then it will automatically take you to the output tab (**Alerts**), where we can see the vulnerabilities present on the given website.



RESULT:

Thus, the features of a Vulnerability Assessment Tool were explored, and a security audit was performed on the target website to identify and document potential vulnerabilities.

EX NO. : 11 A	**DEFEATING MALWARE - BUILDING TROJANS**
DATE :	

AIM:

To build a Trojan and understand the harmful effects of Trojan malware in a computer system.

PROCEDURE:

1. Create a simple Trojan using a Windows Batch File (.bat).
2. Type the following code in Notepad and save it as Trojan.bat.
3. Double-click on the Trojan.bat file.
4. When the Trojan code executes, it will open MS-Paint, Notepad, Command Prompt, Explorer, and other applications infinitely.
5. Restart the computer to stop the execution of this Trojan.

TROJAN:

- In computing, a Trojan horse or Trojan is a malware that misleads users about its true intent.
- Trojans are generally spread through social engineering, such as email attachments disguised as harmless files or by clicking fake advertisements.
- Many modern Trojans act as a backdoor and allow unauthorized access to the affected computer.
- Trojans may allow attackers to access personal information such as banking details, passwords, and identity-related data.
- Example: Ransomware attacks are often carried out using a Trojan.

PROGRAM

Trojan.bat

```
@echo off
:x
start mspaint
start notepad
start cmd
start explorer
start control
start calc
goto x
```

OUTPUT

(MS-Paint, Notepad, Command Prompt, Explorer, and other applications will open infinitely.)

RESULT:

Thus, a Trojan has been built and the harmful effects of Trojan malware have been explored.

EX NO. : 11 B	**DEFEATING MALWARE - ROOTKIT HUNTER**
DATE :	

AIM:

To install and use Rootkit Hunter (RKHunter) in a Linux virtual machine to scan the system and identify possible rootkits or malware.

OBJECTIVE:

- To understand the purpose of Rootkit Hunter (RKHunter).
- To install RKHunter on a Linux system.
- To scan the system for rootkits and malware.
- To identify suspicious files and directories from the scan results.

REQUIREMENTS:

- VirtualBox
- Ubuntu Linux ISO
- Linux Virtual Machine
- Internet connection
- RKHunter tool

PROCEDURE:

Install a virtual machine and boot a Linux OS in a virtual machine and open it.

Virtualbox: https://download.virtualbox.org/virtualbox/7.2.6/VirtualBox-7.2.6a-172322Win.exe

Ubuntu ISO: https://ubuntu.com/download/desktop/thankyou?version=25.10&architecture=amd64

Step 1: Before installing the tool, we should update our Linux System.

Command: sudo apt update Command: sudo apt upgrade –y

This image represents that all updates have been completed.

Step 2: Install the Official RKHUNTER tool for Linux distros.

Command: sudo apt install -y rkhunter

In my computer, I have already installed the rkhunter tool, so it tells that **"rkhunter is already the newest version"**

```
┌──(root㉿ghost)-[/home/ghost]
└─# sudo apt install -y rkhunter
rkhunter is already the newest version (1.4.6-13).
The following packages were automatically installed and are no longer r
  curlftpfs                   libdnnl3.6                  libmpeg2encpp-2.1-
  gccgo-14                    libfuse2t64                 libmplex2-2.1-0t64
  gccgo-14-x86-64-linux-gnu   libgav1-1                   libmupdf25.1
  libaudio2                   libgdk-pixbuf2.0-bin        libonnx1t64
  libavfilter10               libgo-14-dev                libonnxruntime-pro
  libavformat61               libgo23                     libonnxruntime1.21
  libblosc2-4                 libicu76                    libplacebo351
  libconfig-inifiles-perl     libio-pty-perl              libpocketsphinx3
  libcrypt-dev                libipc-run-perl             libpostproc58
  libdlt2                     libmjpegutils-2.1-0t64      librubberband2
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 54
```

Step 3: To run a scan of your system to find rootkits, use this command:
Command: sudo rkhunter –check

```
┌──(root㉿ghost)-[/home/ghost]
└─# sudo rkhunter --check
[ Rootkit Hunter version 1.4.6 ]

Checking system commands ...

  Performing 'strings' command checks
    Checking 'strings' command                          [ OK ]

  Performing 'shared libraries' checks
    Checking for preloading variables                   [ None found ]
    Checking for preloaded libraries                    [ None found ]
    Checking LD_LIBRARY_PATH variable                   [ Not found ]

  Performing file properties checks
    Checking for prerequisites                          [ OK ]
    /usr/sbin/adduser                                   [ OK ]
    /usr/sbin/chroot                                    [ OK ]
    /usr/sbin/cron                                      [ Warning ]
    /usr/sbin/depmod                                    [ OK ]
    /usr/sbin/fsck                                      [ OK ]
    /usr/sbin/groupadd                                  [ OK ]
    /usr/sbin/groupdel                                  [ OK ]
    /usr/sbin/groupmod                                  [ OK ]
    /usr/sbin/grpck                                     [ OK ]
    /usr/sbin/ifconfig                                  [ OK ]
    /usr/sbin/ifdown                                    [ OK ]
    /usr/sbin/ifup                                      [ OK ]
    /usr/sbin/init                                      [ Warning ]
    /usr/sbin/insmod                                    [ OK ]
    /usr/sbin/ip                                        [ OK ]
    /usr/sbin/lsmod                                     [ OK ]
```

The rkhunter tool will start to scan our system for any malware and rootkits.

Step 4: We should press <ENTER> key whenever it asks us. So it will move to next scan.

```
    /usr/bin/gawk
    /usr/bin/lwp-request
    /usr/bin/plocate
    /usr/bin/bsd-mailx
    /usr/bin/dash
    /usr/bin/x86_64-linux-gnu-size
    /usr/bin/x86_64-linux-gnu-strings
    /usr/bin/inetutils-telnet
    /usr/bin/which.debianutils
    /usr/lib/systemd/systemd

[Press <ENTER> to continue]
```

Step 5: Checking for rootkits after <ENTER> key is pressed.

```
Checking for rootkits ...

  Performing check of known rootkit files and directories
    55808 Trojan - Variant A                          [ Not found ]
    ADM Worm                                          [ Not found ]
    AjaKit Rootkit                                    [ Not found ]
    Adore Rootkit                                     [ Not found ]
    aPa Kit                                           [ Not found ]
    Apache Worm                                       [ Not found ]
    Ambient (ark) Rootkit                             [ Not found ]
    Balaur Rootkit                                    [ Not found ]
    BeastKit Rootkit                                  [ Not found ]
    beX2 Rootkit                                      [ Not found ]
    BOBKit Rootkit                                    [ Not found ]
    cb Rootkit                                        [ Not found ]
    CiNIK Worm (Slapper.B variant)                    [ Not found ]
    Danny-Boy's Abuse Kit                             [ Not found ]
    Devil RootKit                                     [ Not found ]
    Diamorphine LKM                                   [ Not found ]
```

Step 6: The **[WARNING]** in red colour tells that, there might be rootkits/ malwares present for that shown directory / folder.

```
  Performing system configuration file checks
    Checking for an SSH configuration file                  [ Found ]
    Checking if SSH root access is allowed                  [ Warning ]
    Checking if SSH protocol v1 is allowed                  [ Not set ]
    Checking for other suspicious configuration settings    [ None found ]
    Checking for a running system logging daemon            [ Found ]
    Checking for a system logging configuration file        [ Found ]

  Performing filesystem checks
    Checking /dev for suspicious file types                 [ Warning ]
    Checking for hidden files and directories               [ Warning ]

[Press <ENTER> to continue]
```

OUTPUT

```
System checks summary
=====================

File properties checks...
    Files checked: 141
    Suspect files: 12

Rootkit checks...
    Rootkits checked : 498
    Possible rootkits: 5

Applications checks...
    All checks skipped

The system checks took: 4 minutes and 38 seconds

All results have been written to the log file: /var/log/rkhunter.log

One or more warnings have been found while checking the system.
Please check the log file (/var/log/rkhunter.log)
```

If we need to see the results after we restart our computer, we can use this command:

Command: *cat /var/log/rkhunter.log*

RESULT:

Thus, the RKHunter tool was successfully installed and executed. The system was scanned for possible rootkits and malware, and any suspicious files or directories were indicated through warning messages.

EX NO. : 12	**NETWORK INTRUSION DETECTION AND EVIDENCE LOGGING**
DATE :	

AIM:

To capture and analyze network traffic using Wireshark in order to detect intrusion activities and log network evidence for forensic analysis.

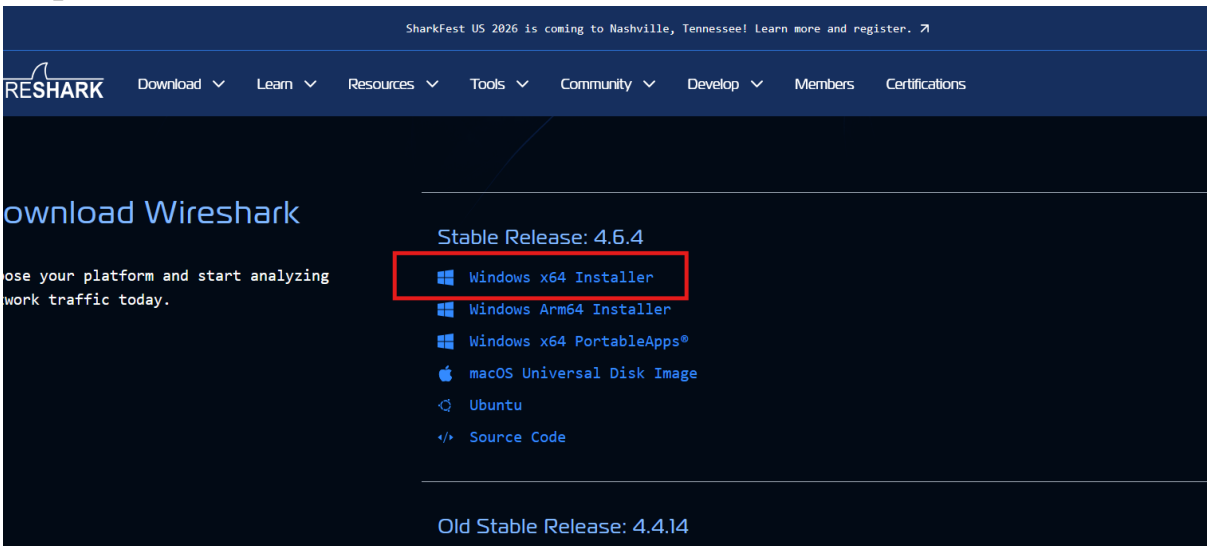
SOFTWARE REQUIREMENTS:

- Wireshark
- Operating System: Windows / Linux
- Active network connection

PROCEDURE:

Step 1: Install Wireshark from the official website (https://www.wireshark.org/download.html)

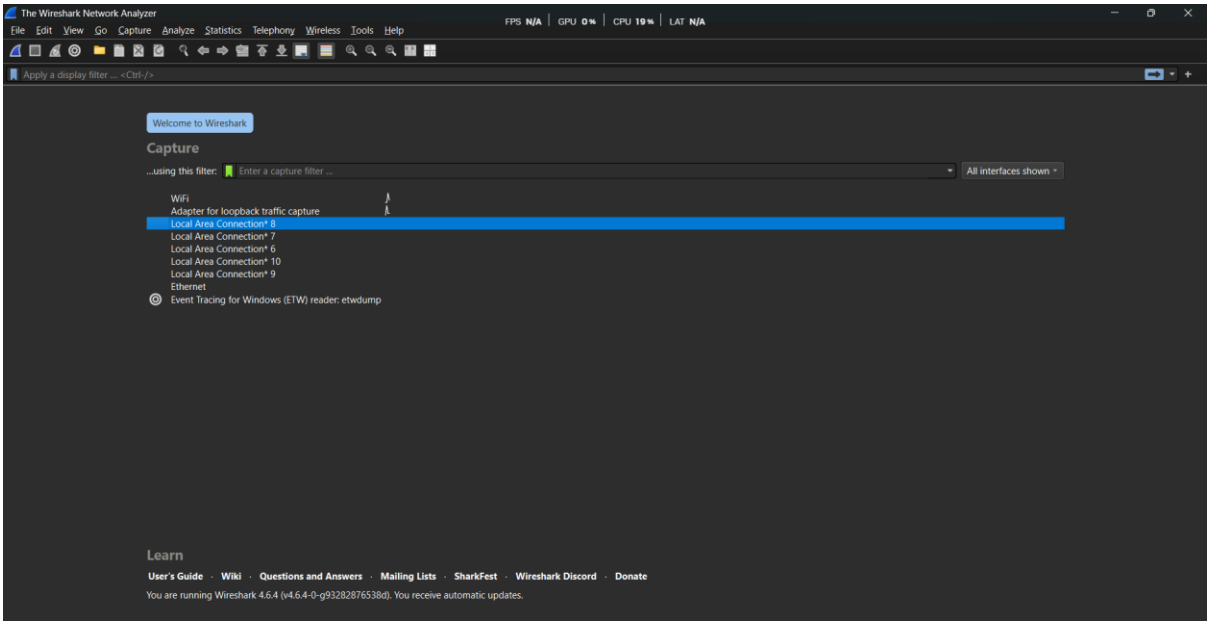
Step 2: Install the Windows x64 installer.



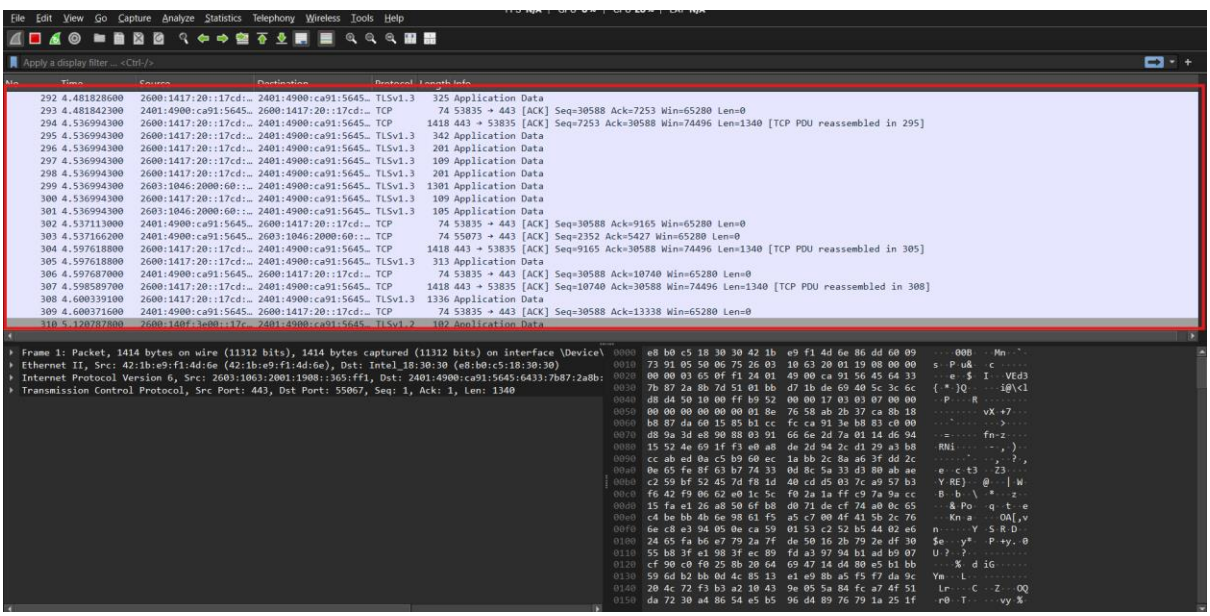
Step 3: Run the downloaded executable file and click "**Next**" for all and install.

Step 4: A reboot is required to efficiently make the Wireshark to capture packets.

Step 5: Open the Wireshark and select the network adapter in which you're using

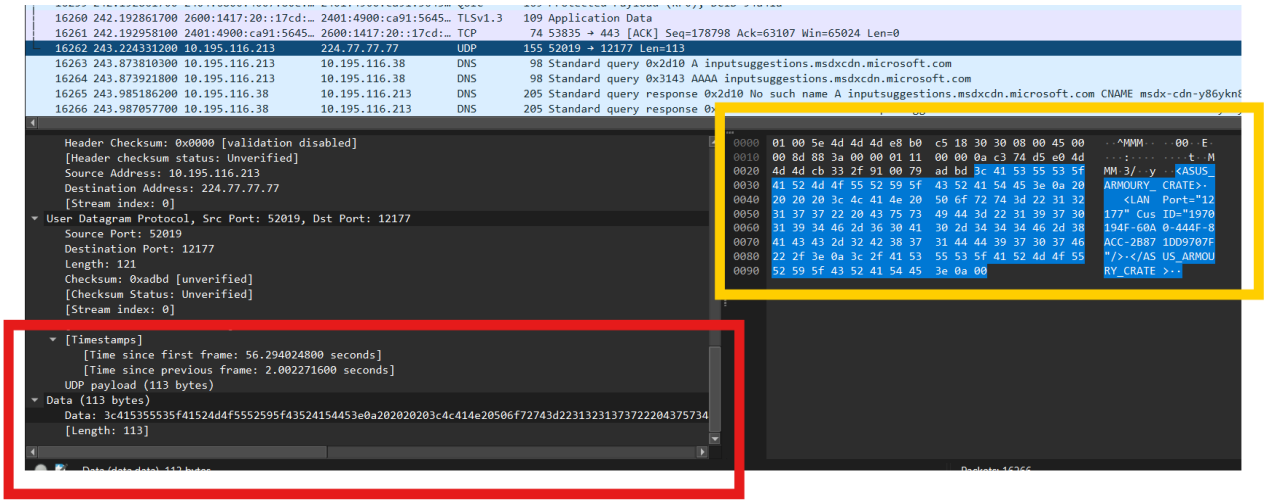


your current internet. Warning: Please select "**wlan0 – wireless LAN" or "wi-fi"** if you use wi-fi and if you use Ethernet , please select **"Ethernet" or "eth0", "eth1"** etc…

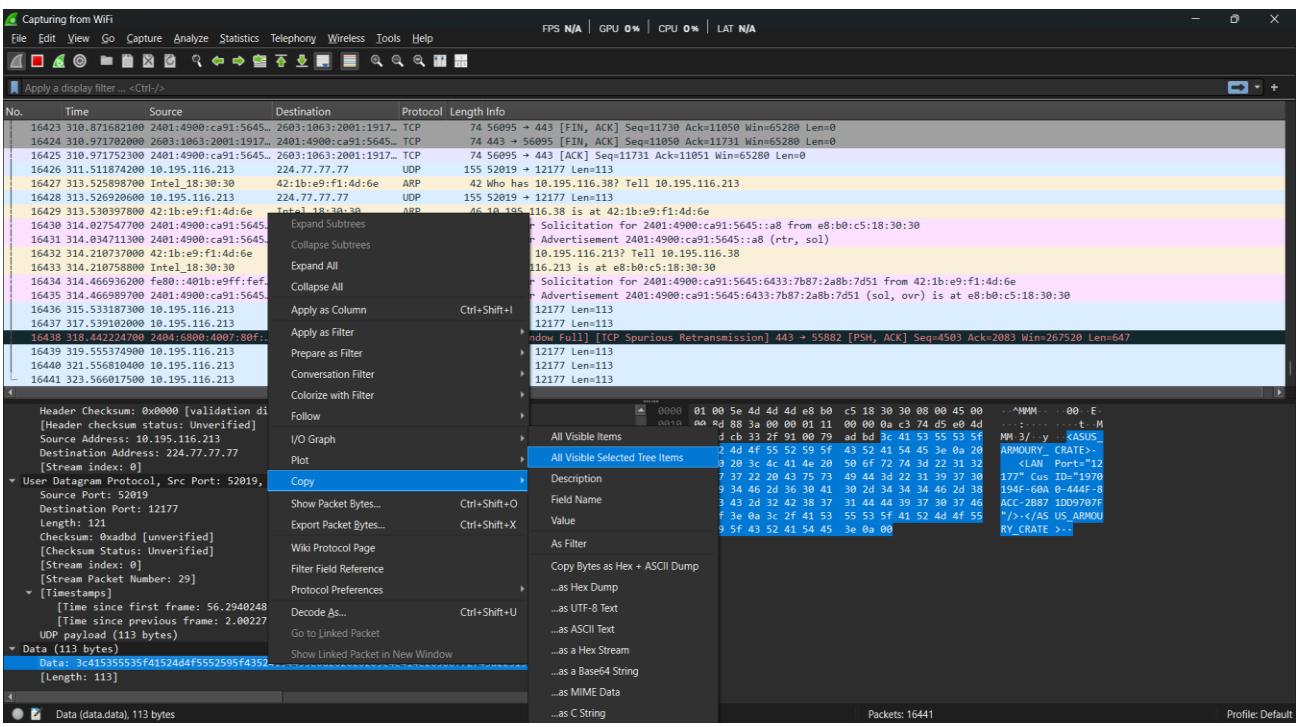


Step 6: You can now able to view all the network packets one by one. You can record and store them or analysis them.

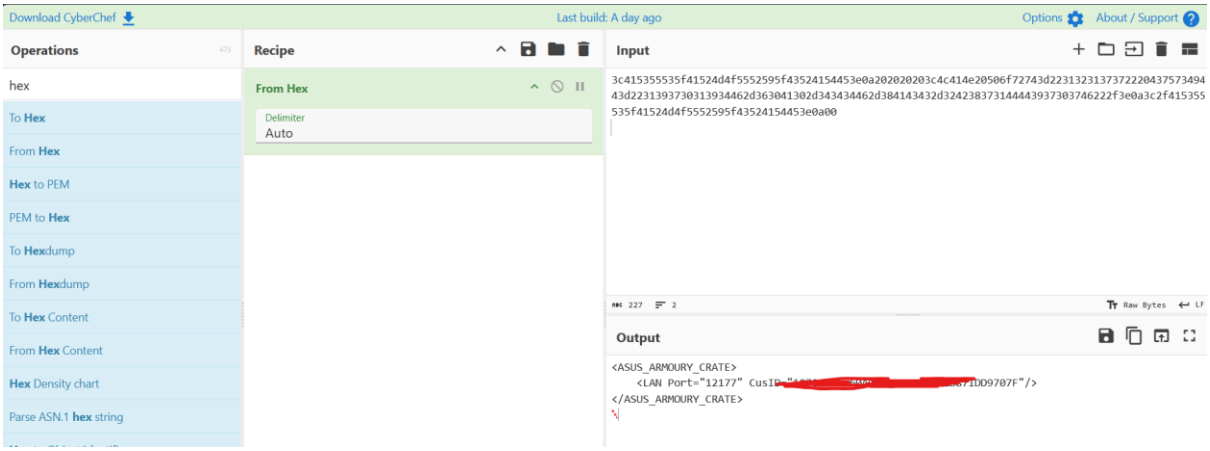
Step 7: The below image shows the encrypted data in the red coloured box of an particular network packet and the yellow colour box represents the hexadecimal dumps with decrypted output at very last.



Step 8: Decoding of the data. Right click the encrypted data to copy it. Then go to copy >> then copy as All selected tree items.



Step 9: Go to Cyberchef or some other website for Hex decoding and enter the copied value to get the decrypted string.



Step 10: We have successfully captured the packet, copied the data from the packet and decrypted it using hex decryptor (Cyber chef)

RESULT:

Wireshark successfully captured and analyzed network traffic, enabling identification of potential intrusion activities. The captured data was logged and preserved as digital evidence for forensic analysis.